

Mike Ichikawa

DATA SCIENTIST · ML ENGINEER · DATA ENGINEER

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SUMMARY

MS Mathematics (CCNY, 2014) and BS Mechanical Engineering (UC Berkeley). Three years in semiconductor Physical Design at Intel; nine years teaching undergraduate mathematics at Santa Rosa Junior College. Over the past eighteen months, a self-directed transition into data science and ML engineering — sixteen production-quality portfolio projects built from scratch across time series forecasting, anomaly detection, NLP, cloud ETL, ML serving, a Databricks medallion lakehouse, and C++ performance extensions via pybind11, plus a complete five-repo paper trading system: live market data ingestion through dual-path signal generation, parameter-optimizing backtesting, and a live Streamlit oversight dashboard.

TECHNICAL SKILLS

Languages	Python · SQL · C++ · R · TypeScript · Node.js
ML / Statistics	scikit-learn · Prophet · PyTorch · XGBoost · LightGBM · scipy · statsmodels · Bayesian inference
LLM / NLP	Anthropic API · Azure OpenAI · Microsoft Graph · MSAL · HuggingFace · FinBERT · prompt engineering
Data Engineering	PostgreSQL · Databricks · Delta Lake · medallion · dbt · SQLAlchemy · pandas
Cloud / Infra	AWS (S3 · Lambda · DynamoDB · CloudWatch) · Docker · FastAPI · Redis · Pydantic v2 · boto3
Tools	Git · pytest · Vitest · Jupyter · Streamlit · Docker Compose

PORTFOLIO PROJECTS — full portfolio at mtichikawa.github.io

Trading System Arc

[Live Demo](#) · github.com/mtichikawa — 5 repos

Five interconnected repositories: live OHLCV pipeline (ccxt/Kraken + PostgreSQL) → candlestick chart generation (mplfinance) → dual-path signal engine (EMA/RSI/MACD/BB + local FinBERT sentiment, 0.6/0.4 fusion) → parameter-optimizing backtester (Sharpe/Sortino/drawdown, staged grid search, T3 feedback loop) → Streamlit multi-page oversight dashboard with live signal display, equity curve, and parameter review. 174/174 tests across T2–T5.

ccxt · PostgreSQL · mplfinance · FinBERT · pandas · NumPy · Streamlit · Plotly · pytest

Databricks Medallion Lakehouse

github.com/mtichikawa/databricks-lakehouse

Medallion lakehouse (bronze → silver → gold) on 10M NYC taxi rows with Delta Lake. Append-only bronze with provenance metadata; silver types, dedupes, and quarantines invalid rows rather than dropping them; gold serves query-specific aggregates. Row-level data quality gates between layers halt the pipeline on schema drift or null-rate breaches. 95 tests across bronze/silver/gold/quality/pipeline. Runs locally via deltalake; deployable to Databricks Community Edition.

Delta Lake · medallion architecture · pandas · pyarrow · data quality gates · pytest

Real-Time Anomaly Detection

github.com/mtichikawa/anomaly-detection

Streaming ensemble detector: Isolation Forest, LSTM autoencoder, and Z-score with 2-of-3 majority voting. Dockerized FastAPI REST API with health checks and configurable detector selection. Live interactive demo on portfolio site.

Docker · FastAPI · scikit-learn · PyTorch · Redis · pytest

Cloud ETL Pipeline

github.com/mtichikawa/cloud-etl-pipeline

AWS-native ETL: tenacity-based REST API ingestion with exponential backoff → Parquet/Snappy storage on S3 with Hive date partitioning → Lambda validation → DynamoDB run logging for failure traceability. Tests mock all AWS calls with moto.

AWS S3 · Lambda · DynamoDB · boto3 · pandas · Apache Parquet · moto

Dockerized ML API

github.com/mtichikawa/dockerized-ml-api

Production ML serving: async thread pool inference, Redis TTL caching (cache hit skips model entirely), Pydantic v2 request validation, async job queue with polling endpoints, Docker Compose deployment. 32 tests.

FastAPI · Docker · Redis · asyncio · Pydantic v2 · scikit-learn

SQL Analytics Pipeline

github.com/mtichikawa/sql-analytics-pipeline

Three-layer dbt-style pipeline (staging → clean → marts) on NYC Airbnb data. Window functions, CTEs, SQLAlchemy ORM with swappable SQLite/PostgreSQL backend. Layered transform pattern enforces raw data immutability.

PostgreSQL · SQLAlchemy · dbt-style transforms · SQLite · pandas

LLM Data Analysis Assistant

github.com/mtichikawa/llm-data-assistant

Two-path query routing: rule-based fast path handles ~70% of queries (correlations, aggregations, group-bys) with zero API cost. Queries below confidence threshold escalate to Anthropic API with DataFrame context injection and multi-turn conversation history.

Anthropic API · pandas · hybrid routing · multi-turn conversation

M365 Enterprise Software (Private)

Full-stack enterprise software for the Microsoft 365 ecosystem, built on Azure. Single-process Node.js backend with Azure OpenAI smart routing (GPT-4o/mini), Microsoft Graph API integration across the full M365 surface, OAuth2/MSAL device-code auth with encrypted per-user token cache, and event-driven webhooks via Graph change notifications. 152 tests (Vitest), TypeScript strict mode, ESLint + Prettier.

TypeScript · Node.js · Azure OpenAI · Microsoft Graph · MSAL · Bot Framework SDK · Vitest

Additional: Multi-Armed Bandit A/B Testing (Thompson Sampling · UCB1 · Bayesian inference) · Bias Detection in LLMs (ANOVA · Cohen's d) · Financial NLP Parser (SEC EDGAR · regex · sentiment lexicon) · GitHub Trend Forecaster (Prophet · PyTorch LSTM in progress)

EXPERIENCE

Mathematics Faculty

2015 – 2023

Santa Rosa Junior College · Santa Rosa, CA

- Taught undergraduate mathematics for nine years: Algebra through Calculus III, Linear Algebra, Differential Equations, Statistics
- Designed and maintained curriculum assessment systems to track student outcomes across multiple cohorts
- Developed data-driven iterative improvements to course materials based on longitudinal performance analysis
- Routinely explained complex mathematical concepts clearly to students with varying preparation levels

Physical Design Engineer

2008 – 2011

Intel Corporation · Semiconductor R&D

- Microprocessor physical design on Intel Ivy Bridge (22nm) — timing closure, power analysis, design rule verification
- Built Python and Perl automation for physical verification workflows, processing millions of layout data points per run
- Identified and flagged yield-risk patterns in pre-tape-out statistical data, contributing to sign-off decisions

EDUCATION

MS Mathematics

The City College of New York

2014

BS Mechanical Engineering

University of California, Berkeley

2008